STAT 6545 – Multivariate Statistical Methods

Multivariate Profiling and Segmentation of Credit Card Customers:
Financial Capacity, Spending Behavior, and Credit Use

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1 Problem Background and Importance

Credit-card customers differ simultaneously in financial capacity, spending intensity, transaction frequency, repayment behavior, credit utilization, digital engagement, and creditworthiness. These characteristics are naturally interrelated. For example, income and credit limit may move together, spending may be associated with both income and available credit, and utilization may be related to outstanding balance and credit score. Consequently, studying one variable at a time can miss important dependence patterns and may provide an incomplete description of customer behavior. The purpose of this project is to use multivariate statistical methods to develop a coherent description of credit-card customer profiles. The project mainly aims at demonstrating the methods discussed in the class and then possibly further outlook based results of common methods and time availability. We set five main goals: (i) reducing a moderately high-dimensional set of customer variables to a smaller number of interpretable dimensions, (ii) identifying and characterizing customer segments, (iii) studying dependence between financial-capacity variables and behavioral variables, (iv) predicting important customer outcomes, and (v) testing whether multivariate behavioral profiles differ across customer groups.

The central research question is:

> *How can multivariate statistical methods be used to characterize, reduce, group, explain, and predict the financial and transactional behavior of credit-card customers?*

Procedures, and analyses aimed at answering the above questions perfectly fits into the purpose of project we are aiming to do in our class as far as the writeup guideline is concerned. Since our data contains multiple continuous, count/discrete, and categorical variables and permits the use of dimension reduction, clustering, discrimination, dependence analysis, regression, and MANOVA within one coherent application.

2 Data Description

The proposed data are from a Kaggle credit-card customer behavior and segmentation dataset. The data file can be accessed at Kaggle: Credit Card Customer Behavior Segmentation (Input). The current data file contains 50,000 customer records and 29 variables describing demographics, financial status, credit-card characteristics, transaction behavior, payment behavior, rewards activity, and digital engagement. `Customer_ID` is an identifier and will not be used as an analytical variable.

For now we have limited the analysis to exploratory analysis only to a correlation matrix for selected continuous variables and histograms for the numeric variables, just to have a glimpse. The full project will use a larger subset of variables after screening for redundancy, scale differences, missingness, and severe numerical diagnostics and detailed outlooks.

Variables to be considered

Variable	Role/type	Brief description
Customer_ID	Identifier	Unique customer identifier; excluded from modeling.
Age	Continuous	Customer age.
Gender	Categorical	Customer gender.
Occupation	Categorical	Customer occupation category.
Card_Type	Categorical	Type or category of credit card.
Annual_Income	Continuous	Annual customer income.
Credit_Limit	Continuous	Total approved credit limit.
Card_Age_Months	Continuous/discrete	Length of time the card has been held, in months.
Monthly_Spending	Continuous	Total monthly credit-card spending.
Monthly_Transactions	Count	Number of transactions per month.
Average_Transaction_Value	Continuous	Average value of a transaction.
Online_Shopping_Spending	Continuous	Monthly spending on online shopping.
Grocery_Spending	Continuous	Monthly grocery spending.
Fuel_Spending	Continuous	Monthly fuel spending.
Dining_Spending	Continuous	Monthly dining spending.
Travel_Spending	Continuous	Monthly travel spending.
Entertainment_Spending	Continuous	Monthly entertainment spending.
Utility_Bill_Spending	Continuous	Monthly utility-bill spending.
Outstanding_Balance	Continuous	Current unpaid balance.
Statement_Balance	Continuous	Balance appearing on the billing statement.
Payment_Amount	Continuous	Amount paid toward the credit-card balance.
Payment_Ratio	Continuous	Payment relative to the amount due/balance.
Credit_Utilization	Continuous	Proportion of available credit currently used.
Cash_Advance_Amount	Continuous	Amount of cash advances.
EMI_Count	Count	Number of EMI/installment transactions or plans.
International_Transactions	Count	Number of international transactions.
Reward_Points_Earned	Continuous/count	Reward points earned.
Reward_Points_Redeemed	Continuous/count	Reward points redeemed.
Mobile_App_Logins	Count	Number/frequency of mobile-app logins.
Credit_Score	Continuous	Customer credit score.

3 Preliminary Exploratory Analysis

First, we used Pearson correlation matrix to have a glimpse of pairwise linear dependence among a selected set of financial and behavioral variables and then, histograms are used to explore the marginal distributional shape, skewness, extreme values, and the discrete nature of count variables.

The initial correlation display already suggests substantial redundancy among several financial variables. In particular, the preliminary plot indicates very strong positive relationships among `Annual_Income`, `Credit_Limit`, and `Monthly_Spending`, together with a strong association between `Credit_Limit` and `Outstanding_Balance`. It also suggests a notable negative relationship between `Credit_Utilization` and `Credit_Score`. These patterns motivate the planned use of principal component/factor methods and careful screening for multicollinearity before regression and discriminant analyses.

The histograms indicate that several monetary variables are positively skewed, whereas some variables such as age, card age, transaction frequency, and mobile-app activity have more regular or approximately symmetric shapes. These features will be considered when standardizing variables and when assessing multivariate normality in the final project. At the proposal stage, however, no formal transformation, outlier removal, or model selection will be performed.

3.1 Data preparation

It common incidence that large data sets have a lot of impurities that can seriously affect the analyses and inferences drown from the data. So, be any further analysis we deep dived into exploration to check for the presence of outlier, missing values, data anomalies(severe deviation from normality) and other form of undesirable properties. Based on outcome we might delete cases, transform columns, scale specific column or even drop column if there exists near linear relationship between variables, that might cause trouble for PCA, CA and other MV methods.

Normality and unusual observations. Following chapter 4 slide, we will examine marginal histograms and normal Q–Q plots, Q–Q plots of principal-component linear combinations, pairwise scatterplots, and chi-square plots for selected subsets and the full retained continuous vector. For a nonsingular p-variable subset, squared distances will be computed as

$$d_i^2 = (\mathbf{x}_i - \bar{\mathbf{x}})^\mathsf{T} S^{-1} (\mathbf{x}_i - \bar{\mathbf{x}}),$$

and their ordered values compared graphically with χ_p^2 quantiles. This is an approximate distributional diagnostic, not an exact chi-square test of normality. Unusual records will be investigated and analyses compared with and without influential observations; legitimate customers will not be deleted solely for being extreme. For MANOVA and discrimination, normality and covariance assumptions will be examined within card types; regression assumptions will be assessed through residuals. Scatterplot matrices, required by the project guidelines, will supplement the existing figures when the customer-level data are processed.

4 Proposed methods of analysis

As the purpose of this project to demonstrate the multivariate method discussed in class, applied to real data *plus some recent MV methods depending on the outcome of the conventional MV methods.* After data cleaning and pre-processing are done we will apply all the methods discussed in lecture mainly chapter 5 through chapter 12 of the slides.

4.1 MV modeling, MV regression (Linear), MANOVA(one-way, two-way, including mean separations)

We will use multivariate regression methods to examine how customer characteristics relate jointly to selected outcomes. Multiple linear regression will be used for continuous outcomes, and logistic regression for binary outcomes. One-way and two-way MANOVA will test whether the mean vectors of selected financial and behavioral variables differ across customer groups, while accounting for multiple outcomes. When a MANOVA indicates group differences, follow-up comparisons will identify which groups or outcomes contribute to the differences— Tyler and Thao.

4.2 Principal component analysis

We will do both Covariance based PCA and Correlation based PCA, with further extension to Kernel PCA and Sparse PCA based on the outcomes of covariance based and correlation based PCA, with all numerical and graphical methods such as scree plot and cumulative covariance plots included— Dejene.

4.3 Factor Analysis: Exploratory Factor Analysis (EFA) and Confirmatory Factor analysis

Exploratory factor analysis (EFA) will be used to identify underlying factors that summarize relationships among correlated customer variables. Confirmatory factor analysis (CFA) will then assess how well a proposed factor structure fits the data.—- Dejene

4.4 Cluster Analysis

We will compare density-based, hierarchical, distribution-based, and grid-based clustering methods to group customers with similar financial and spending profiles, with possible extension to spectral clustering, fuzzy clustering .—Thao

4.5 Canonical Correlation Analysis

Canonical correlation analysis will examine the strength of association between two sets of customer characteristics, such as financial-capacity variables and spending-behavior variables, with possible extension to redundancy analysis, canonical loading and cross loading.–Thao

4.6 Discriminant Analysis

Linear and quadratic discriminant analysis will be used to classify customers into predefined groups, such as card types, based on their financial and behavioral characteristics.–Tobe

4.7 Tree-Based Regression and Predictive Comparison, Logistic regression

To examine nonlinear relationships and interactions that may not be captured by multiple linear regression, we will compare a linear regression model with a regression tree, random forest, and, if appropriate, an extreme gradient boosting model (XGBoost). A continuous outcome such as `Monthly_Spending` or `Credit_Score` will be selected based on data quality and the project question. Predictors will be chosen to avoid using variables that directly define or duplicate the response.

Predictive performance will be evaluated using cross-validation and measures such as root mean squared error (RMSE), mean absolute error (MAE), and coefficient of determination (R^2). We will use variable-importance summaries and partial-dependence plots to describe influential predictors and possible nonlinear patterns. These models will supplement the regression and other multivariate analyses by assessing whether flexible tree-based methods improve out-of-sample prediction.–Tobe

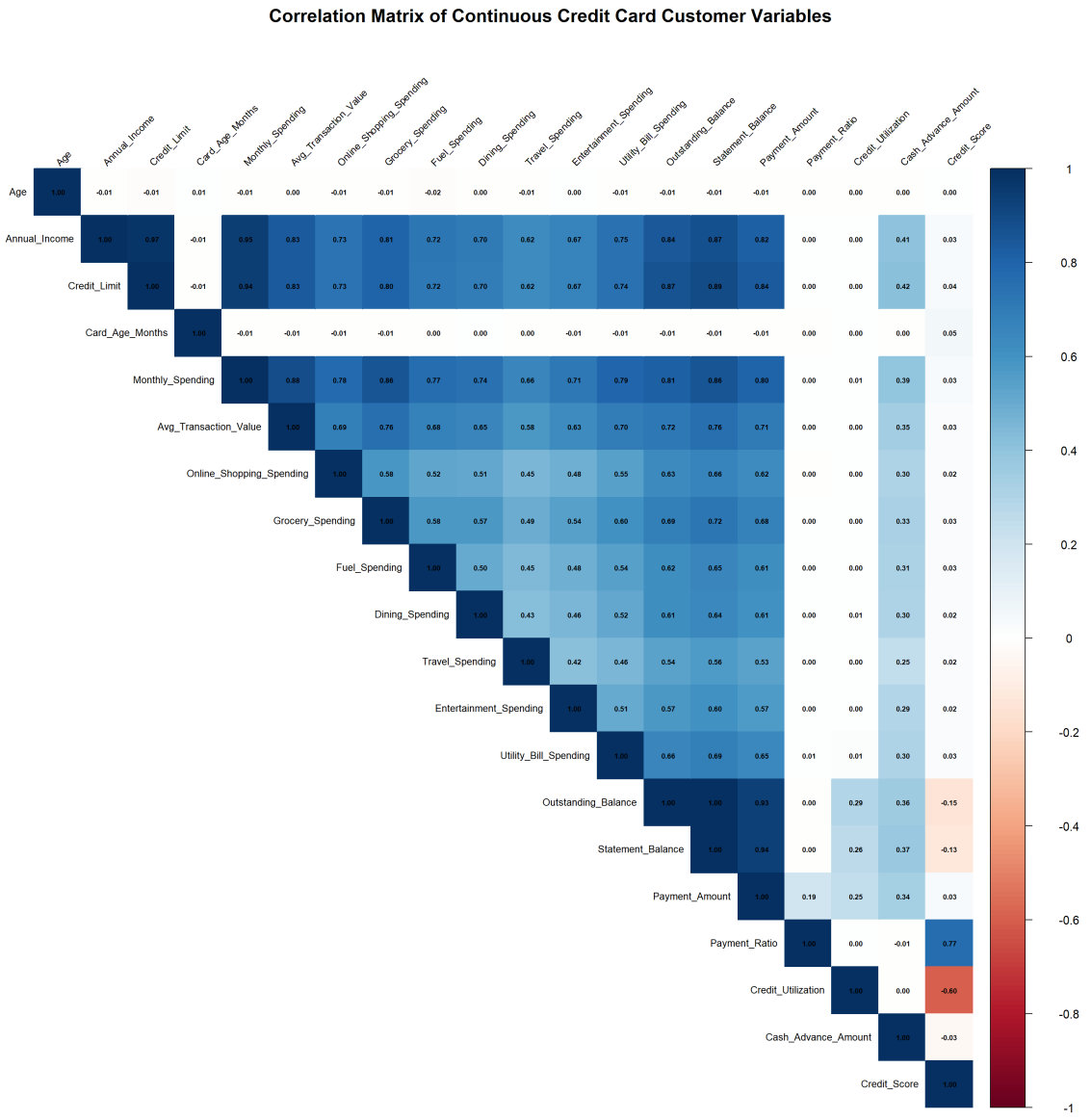


Figure 1: Correlation matrix for selected credit-card customer variables.

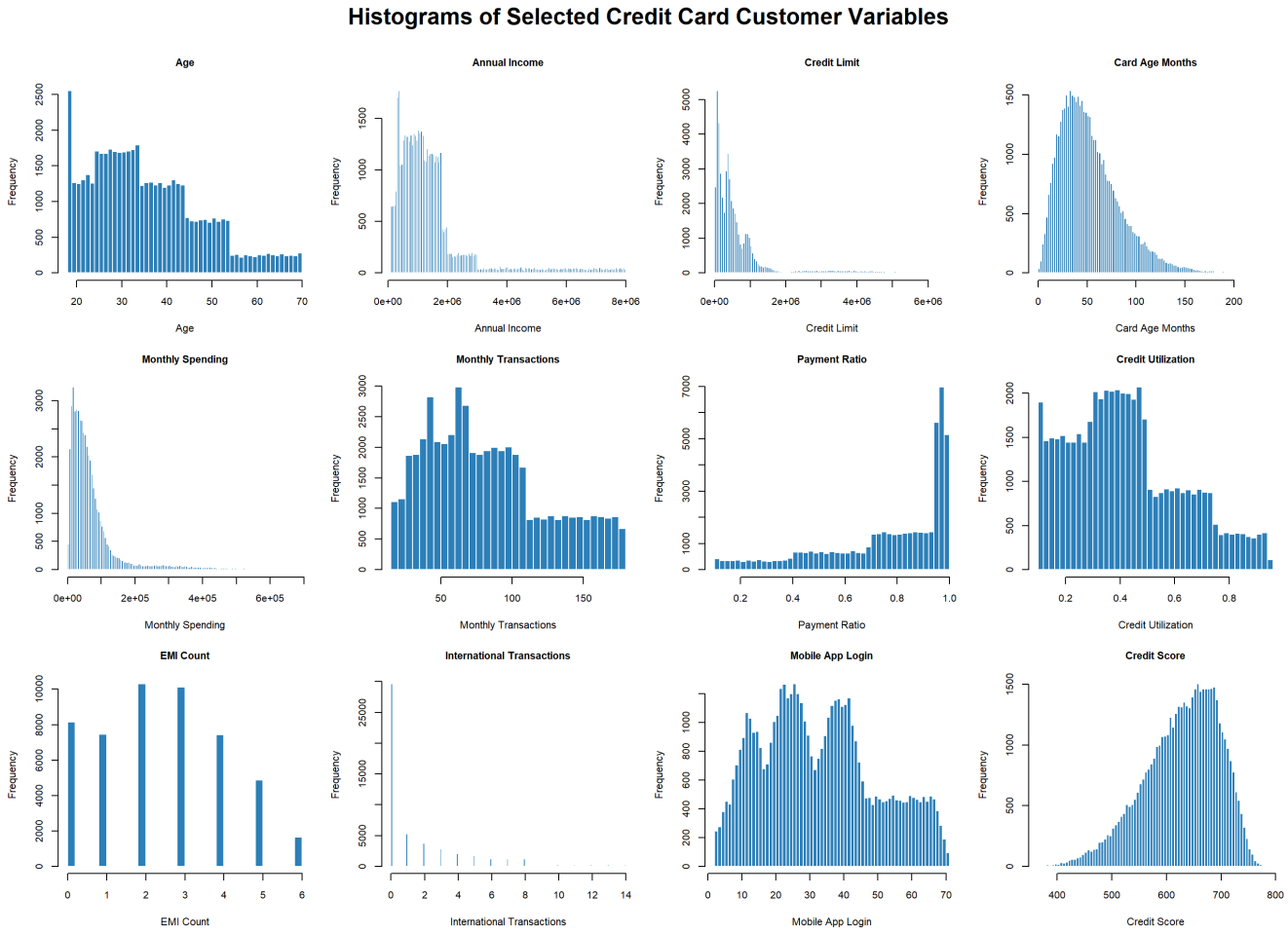


Figure 2: Histograms of the numeric credit-card customer variables.